ODAG Report

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Overview

This analysis seeks to evaluate the relationship between an African country's official development assistance gross (ODAG) and their democracy score. ODAG measures the aid received by an African country from external countries and international organizations (IO). (Note: In this report, we refer to ODAG received from other countries as ODAG and ODAG received from entities that are not countries are ODAGIO.) We used democracy scores from Freedom House (TOTLIB1), the POLITY IV database (POLITY2), and the Liberal Democracy Index from the V-Dem Institute (V2x_Libdem). (Note: Here we only analyze POLITY2, whereas in previous reports we have analyzed both POLITY and POLITY2.)

This main analyses were split into a reduced and full model. The reduced model only considers democracy as a fixed effect, while including random intercepts for the African country and year. These random effects account for structural correlation between observations within the data structure.

The full model includes all of the parts of the reduced model while adding the African country's population, gross national income (GNI), total imports, total exports, colonial Past, region, and a cold war indicator variable. We applied a log transformation to GNI, population, imports, and exports due to their right skew.

ODAG (Countries)

In the reduced models (Tables 1-3), all V2x_Libdem, POLITY2, and TOTLIB1 coefficients are significant at $\alpha = 0.05$. In all cases, higher democracy scores were associated with larger levels of total ODAG.

In the full models (Table 4-6), all V2x_Libdem, POLITY2, and TOTLIB1 coefficients are significant at $\alpha = 0.05$. In all cases, higher democracy scores were associated with high total ODAG. In addition, we see that total population, GNI, total imports, and the cold war indicator are all significant. We also see total exports are significant for most of the different lagged

models for v2x_Libdem models. Population and total imports were positively correlated with total ODAG whereas larger GNI was related to lower levels of ODAG. Finally, the cold war (pre-1992) is associated with lower total levels of ODAG.

ODAG (IO)

In the reduced models (Tables 7-9), once again all V2x_Libdem, POLITY2, and TOTLIB1 coefficients are significant at $\alpha = 0.05$. In all cases, higher democracy scores were associated with larger levels of total ODAG.

In the full models (Table 10-12), all V2x_Libdem, POLITY2, and TOTLIB1 coefficients are again significant at $\alpha = 0.05$. In all cases, higher democracy scores were associated with high total ODAG. As before, we see that total population, GNI, total imports, and the cold war indicator are all significant. Total exports in also significant here. Population and total imports were positively correlated with total ODAG whereas larger GNI and total exports was related to lower levels of ODAG. Finally, the cold war indicator (pre-1992) is associated with lower total levels of ODAG. Also of interest, Italian colonial past is significant in the POLITY2 model, but not in any other models.

Models

Reduced Model

The reduced model is described below, where the response Y_{it} is ODAG for African country i in year t. We applied a log transformation to ODAG for better model fit. The predictor X_{1it} represents the democracy score of African country i in year t.

In lagged models, the year t changes units depending on the lag. For instance, a one-year lag would produce the model equation $\log(\text{ODAG}_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$, where $X_{1i(t-1)}$ is the external country's democracy score in the prior year (year t-1).

$$Y_{it} \sim \text{N}(\mu_{it}, \sigma^2)$$

$$\log(\text{ODAG}_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$$

$$\gamma_{0i} \sim \text{N}(0, \sigma^2_{\text{country}})$$

$$\gamma_{0t} \sim \text{N}(0, \sigma^2_{\text{year}})$$

Full Model

Our full model equations are given below. They follow the same setup as the reduced model with the addition of the aforementioned transformed country characteristics. In lagged models, we only lagged the country's democracy score. For instance, a one-year lag would produce the model equation $\log(\text{ODAG}_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + ... + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$, where $X_{1i(t-1)}$ is the country's democracy score in the prior year (year t-1).

Additional predictors in this model include the African country's log-transformed GNI, population, and total ODAG. Quadratic terms were included for the log-transformed GNI, population, and total ODAG. The log-transformed population variable added a cubic term to capture further nonlinear tendencies. In addition, there are indicator variables for region, colonial past, cold war, and cold war interacted with ideology (i.e., pre-cold war ideology).

$$Y_{it} \sim \text{N}(\mu_{it}, \sigma^2)$$

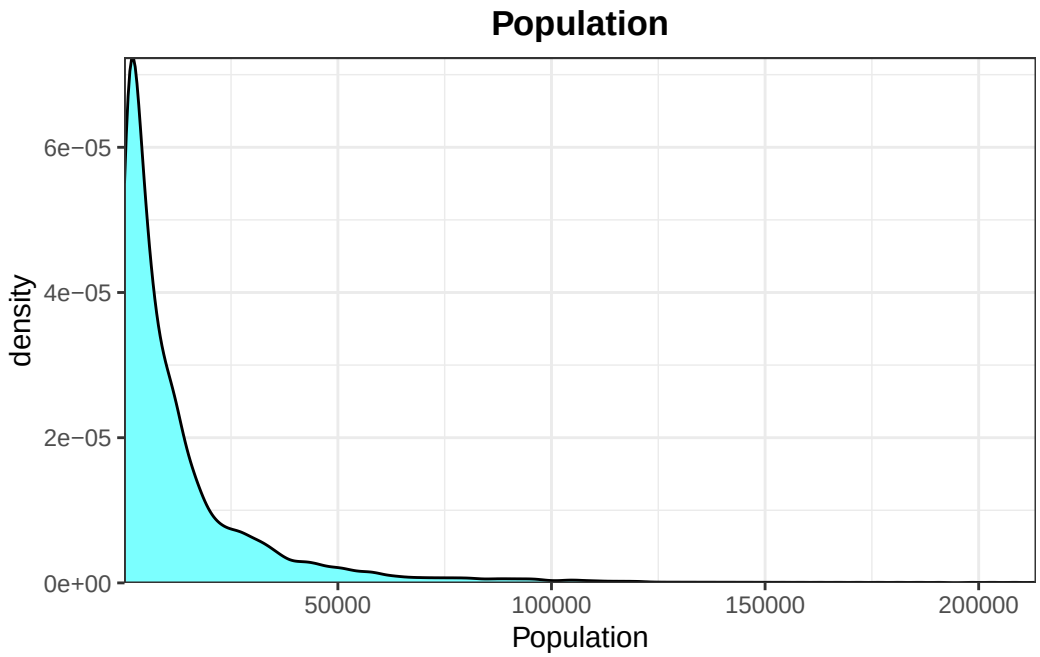
$$\log(\text{ODAG}_{it}) = \beta_0 + \beta_1 X_{1ik} + ... + \gamma_{0i} + \gamma_{tk} + \epsilon_{it}$$

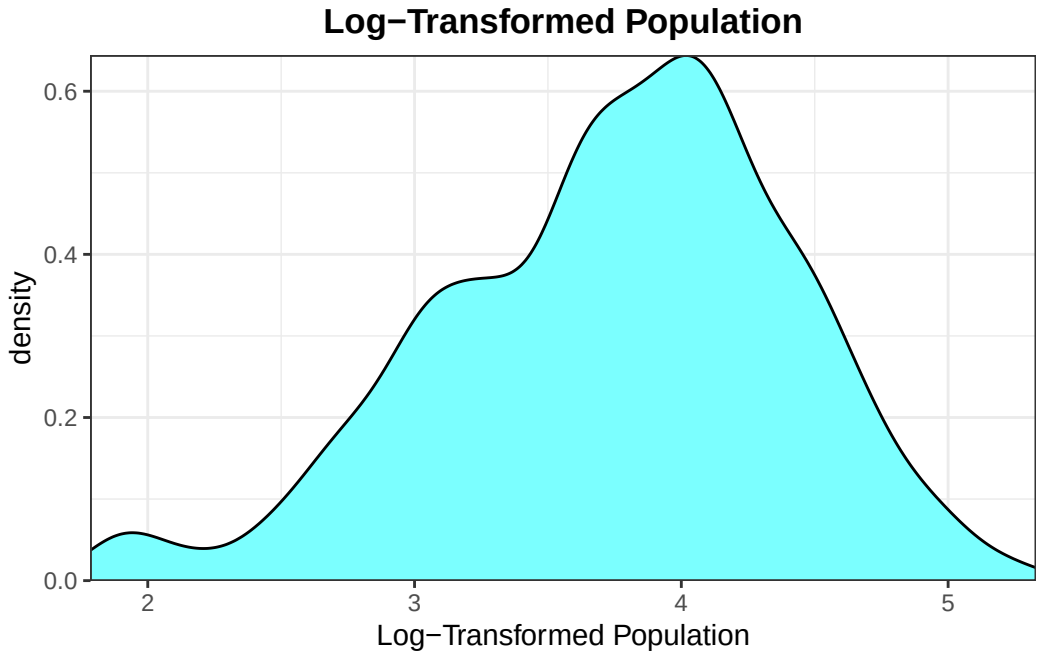
$$\gamma_{0i} \sim \text{N}(0, \sigma^2_{\text{country}})$$

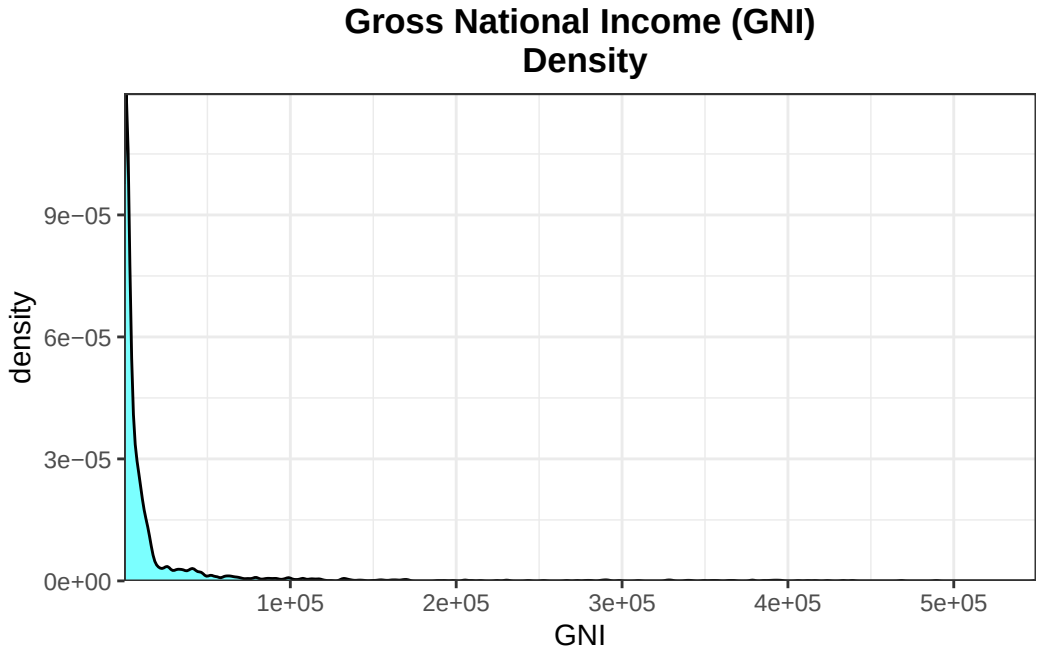
$$\gamma_{0t} \sim \text{N}(0, \sigma^2_{\text{year}})$$

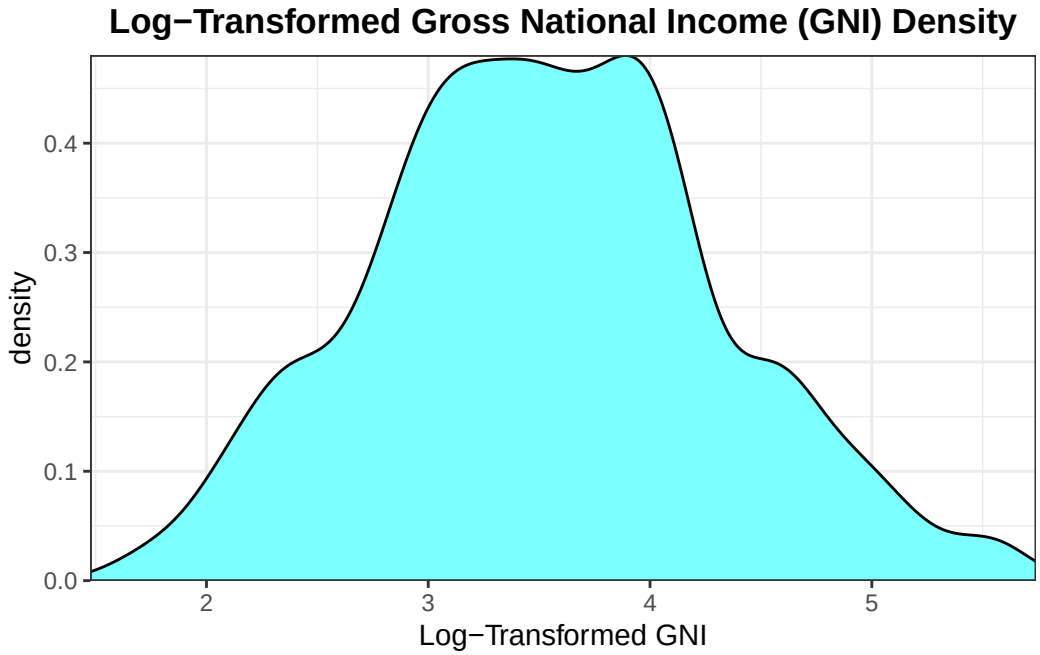
Data Visualizations and Results

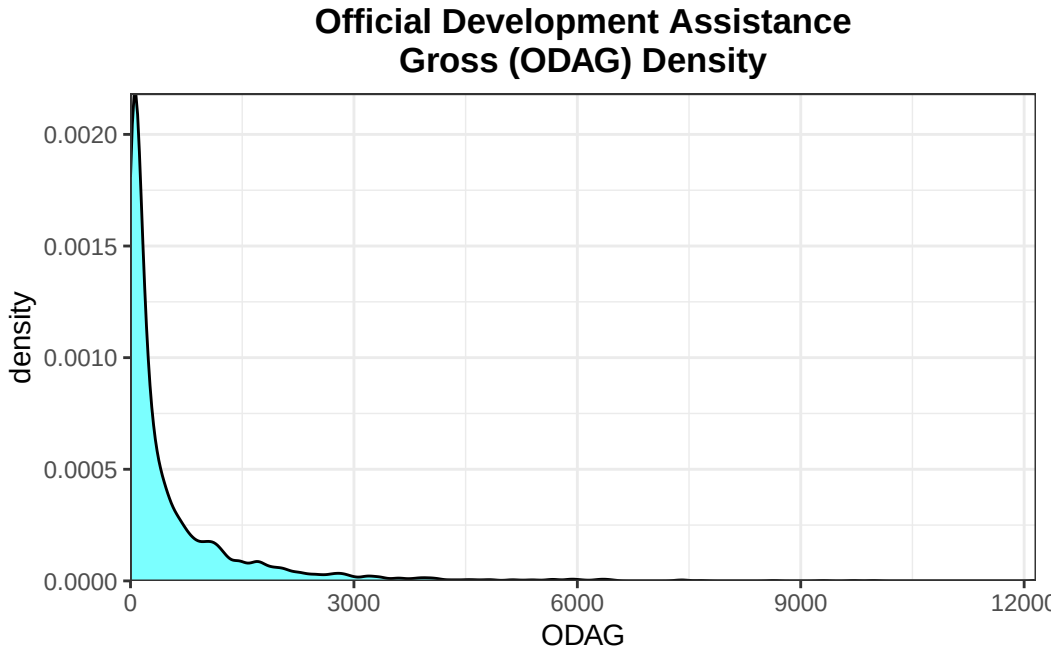
The distributions of the full model predictor variables and their log transformations are plotted below. The log-transformed variables all display a noticeably reduced right-skew, improving their performance and interpretation in the full models.

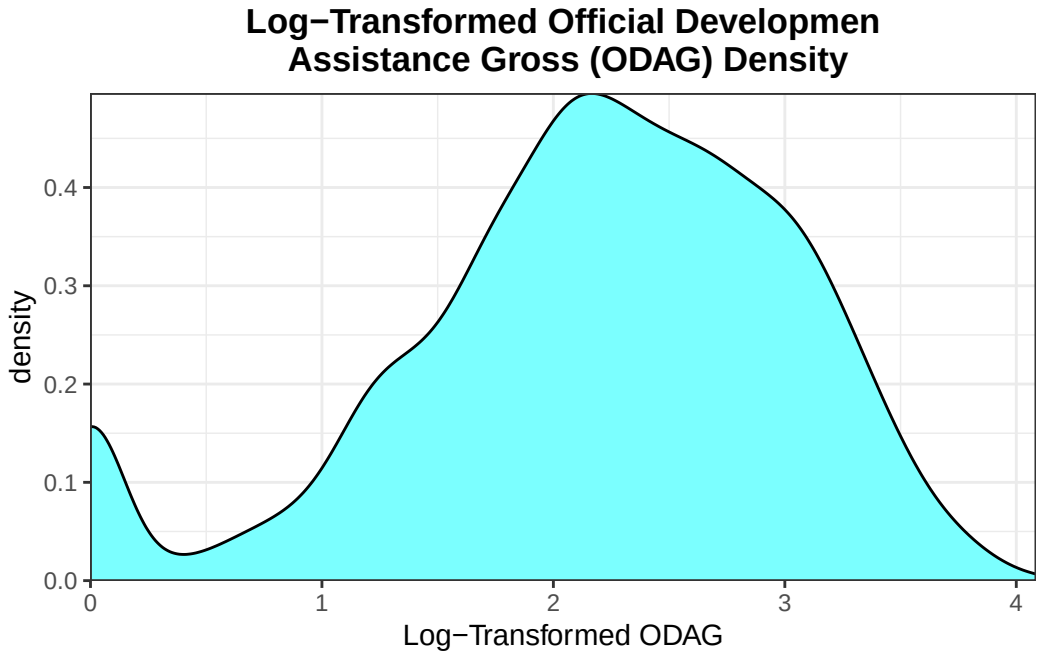

Population
6e−05
4e−05
2e−05
0e+00
density
50000
100000
150000
200000
Population

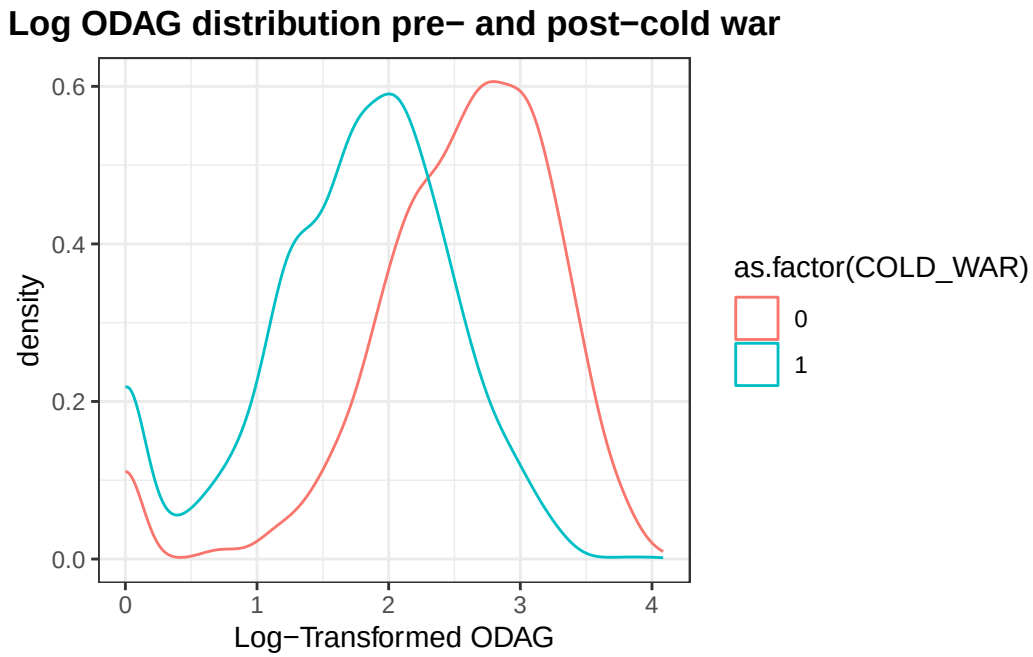

Log−Transformed Population
0.6
0.4
0.2
0.0
density
2
3
4
5
Log−Transformed Population

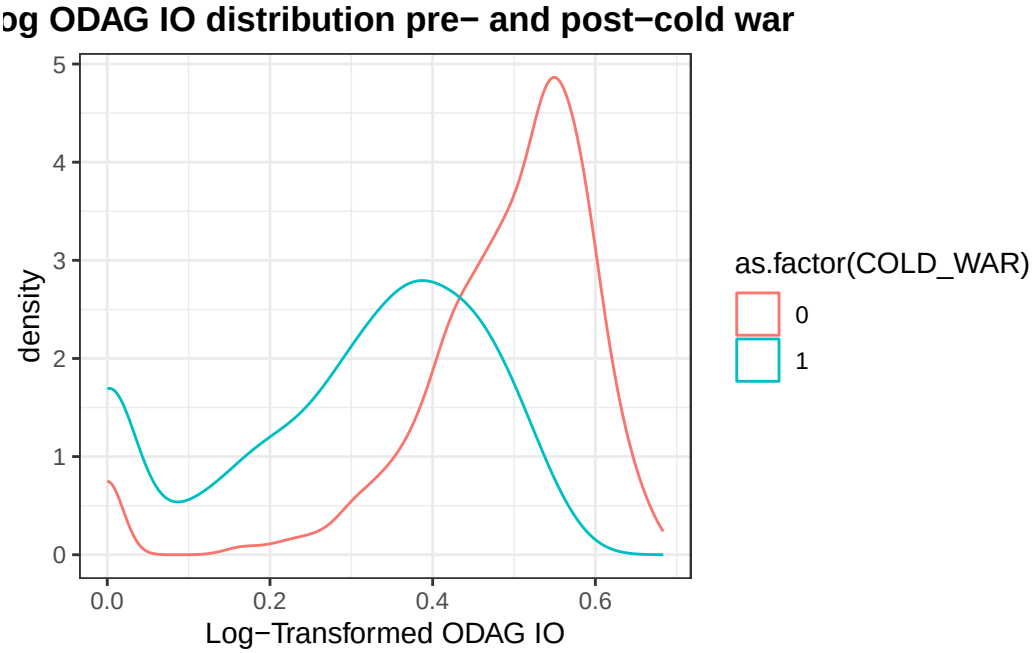

Gross National Income (GNI)
Density
density
9e−05
6e−05
3e−05
0e+00
1e+05
2e+05
3e+05
4e+05
5e+05
GNI


Log−Transformed Gross National Income (GNI) Density
density
0.4
0.3
0.2
0.1
0.0
2
3
4
5
Log−Transformed GNI


Official Development Assistance
Gross (ODAG) Density
density
0.0020
0.0015
0.0010
0.0005
0.0000
0
3000
6000
9000
12000
ODAG


Log−Transformed Official Developmen
Assistance Gross (ODAG) Density
density
0.4
0.3
0.2
0.1
0.0
0
1
2
3
4
Log−Transformed ODAG





Summaries for the reduced and full models for ODAG are shown with the regression coefficient estimates in the column labeled "Est." starred by significance as denoted in the legend below. Next to each coefficient estimate is their respective standard error.

Table 1: **ODAG Countries: Reduced Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	1.491***	0.115	1.499***	0.113	1.527***	0.111	1.559***	0.108
SD (Intercept YEAR)	0.790		0.780		0.748		0.710	
SD (Intercept CCODE)	0.506		0.500		0.504		0.507	
SD (Observations)	0.336		0.333		0.328		0.324	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

ODAG Countries Results

ODAG IO Results

Below are the summaries for the full models for ODAG from IO organizations As with the reduced model summaries, the $\hat{\beta}$ coefficients are starred by significance as denoted in each table, with their respective standard error below.

ODAG Total Results

Table 2: **ODAG Countries: Reduced Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	1.656***	0.114	1.700***	0.112	1.742***	0.109	1.786***	0.106
POLITY2	0.008***	0.002						
POLITY2_lag1			0.007***	0.002				
POLITY2_lag2					0.006***	0.002		
POLITY2_lag3							0.005**	0.002
SD (Intercept YEAR)	0.812		0.784		0.756		0.721	
SD (Intercept CCODE)	0.434		0.437		0.441		0.442	
SD (Observations)	0.334		0.326		0.322		0.318	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 3: **ODAG Countries: Reduced Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	2.453***	0.096	2.474***	0.096	2.468***	0.096	2.462***	0.096
TOTLIB1	−0.034***	0.003						
TOTLIB1_lag1			−0.035***	0.003				
TOTLIB1_lag2					−0.033***	0.003		
TOTLIB1_lag3							−0.031***	0.003
SD (Intercept YEAR)	0.424		0.414		0.407		0.403	
SD (Intercept CCODE)	0.516		0.518		0.517		0.517	
SD (Observations)	0.317		0.315		0.316		0.317	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 4: **ODAG Countries: Full Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−0.179	0.297	−0.229	0.301	−0.260	0.300	−0.222	0.295
logPOP	0.703***	0.062	0.702***	0.063	0.694***	0.063	0.691***	0.062
logGNI	−0.146***	0.032	−0.142***	0.031	−0.136***	0.031	−0.144***	0.031
logTRDIMtot	0.125***	0.020	0.137***	0.020	0.149***	0.020	0.141***	0.021
logTRDEXtot	−0.034+	0.019	−0.040*	0.019	−0.043*	0.019	−0.030	0.020
COLD_WAR	−0.397***	0.084	−0.380***	0.081	−0.359***	0.078	−0.351***	0.073
CW_Socialist	−0.038	0.035	−0.017	0.035	−0.010	0.035	−0.011	0.035
CW_Marxist	0.005	0.037	0.001	0.036	−0.004	0.036	−0.013	0.036
COL_British	−0.103	0.105	−0.108	0.107	−0.115	0.107	−0.120	0.105
COL_Portuguese	0.062	0.146	0.062	0.149	0.057	0.150	0.056	0.147
COL_Spanish	−0.297	0.292	−0.298	0.297	−0.308	0.299	−0.329	0.293
COL_Italian	−0.168	0.215	−0.146	0.219	−0.136	0.220	−0.136	0.216
COL_Belgian	0.074	0.296	0.055	0.302	0.036	0.304	0.010	0.298
COL_German	0.001	0.134	−0.005	0.137	−0.014	0.138	−0.016	0.135
COL_US	0.143	0.280	0.145	0.285	0.140	0.287	0.118	0.281
COL_None	−0.168	0.297	−0.173	0.302	−0.169	0.304	−0.157	0.298
COL_Unique	0.324	0.301	0.319	0.306	0.318	0.307	0.320	0.302
REG_E_Africa	−0.038	0.158	−0.016	0.161	0.000	0.162	−0.002	0.159
REG_S_Africa	−0.187	0.170	−0.165	0.173	−0.143	0.174	−0.139	0.170
REG_C_Africa	−0.167	0.175	−0.141	0.178	−0.115	0.179	−0.111	0.175
REG_W_Africa	−0.251+	0.150	−0.229	0.153	−0.208	0.154	−0.203	0.151
SD (Intercept YEAR)	0.303		0.294		0.278		0.261	
SD (Intercept CCODE)	0.264		0.269		0.271		0.265	
SD (Observations)	0.284		0.279		0.274		0.272	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 5: **ODAG Countries: Full Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.221	0.332	0.192	0.327	0.211	0.319	0.283	0.308
POLITY2	0.008***	0.002						
POLITY2_lag1			0.007***	0.002				
POLITY2_lag2					0.006***	0.002		
POLITY2_lag3							0.005**	0.002
logPOP	0.637***	0.073	0.637***	0.072	0.612***	0.071	0.598***	0.069
logGNI	−0.129***	0.034	−0.133***	0.033	−0.125***	0.033	−0.137***	0.033
logTRDIMtot	0.124***	0.022	0.139***	0.022	0.156***	0.022	0.148***	0.022
logTRDEXtot	−0.020	0.021	−0.025	0.021	−0.031	0.021	−0.014	0.021
COLD_WAR	−0.490***	0.088	−0.480***	0.083	−0.459***	0.077	−0.453***	0.071
CW_Socialist	−0.036	0.037	−0.013	0.037	0.012	0.037	0.022	0.037
CW_Marxist	−0.014	0.039	−0.020	0.038	−0.024	0.038	−0.033	0.037
COL_British	−0.084	0.108	−0.088	0.107	−0.086	0.106	−0.078	0.103
COL_Portuguese	0.100	0.161	0.090	0.160	0.074	0.157	0.069	0.153
COL_Spanish	−0.388	0.294	−0.413	0.293	−0.450	0.288	−0.491+	0.280
COL_Italian	−0.315	0.215	−0.291	0.214	−0.279	0.211	−0.276	0.205
COL_Belgian	0.062	0.297	0.042	0.295	0.037	0.291	0.026	0.282
COL_German	0.045	0.134	0.036	0.133	0.031	0.131	0.033	0.127
COL_US	0.148	0.278	0.139	0.277	0.126	0.272	0.094	0.264
COL_None	−0.184	0.297	−0.186	0.295	−0.162	0.291	−0.128	0.282
COL_Unique	0.253	0.308	0.260	0.303	0.324	0.299	0.353	0.288
REG_E_Africa	−0.083	0.159	−0.056	0.158	−0.040	0.155	−0.040	0.151
REG_S_Africa	−0.191	0.174	−0.163	0.173	−0.142	0.170	−0.140	0.165
REG_C_Africa	−0.209	0.177	−0.182	0.176	−0.158	0.174	−0.154	0.168
REG_W_Africa	−0.235	0.152	−0.210	0.151	−0.188	0.148	−0.182	0.144
SD (Intercept YEAR)	0.310		0.294		0.269		0.247	
SD (Intercept CCODE)	0.261		0.260		0.256		0.249	
SD (Observations)	0.294		0.289		0.285		0.282	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 6: **ODAG Countries: Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.335	0.290	0.285	0.293	0.296	0.289	0.307	0.284
TOTLIB1	−0.034***	0.003						
TOTLIB1_lag1			−0.036***	0.003				
TOTLIB1_lag2					−0.034***	0.003		
TOTLIB1_lag3							−0.032***	0.003
logPOP	0.728***	0.064	0.734***	0.065	0.721***	0.065	0.706***	0.064
logGNI	−0.212***	0.034	−0.207***	0.034	−0.200***	0.034	−0.194***	0.035
logTRDIMtot	0.132***	0.021	0.157***	0.022	0.155***	0.022	0.147***	0.023
logTRDEXtot	0.011	0.020	−0.012	0.021	−0.010	0.022	−0.001	0.022
COLD_WAR	−0.218***	0.053	−0.187***	0.051	−0.167***	0.048	−0.152**	0.046
CW_Socialist	0.020	0.043	0.032	0.043	0.025	0.044	0.019	0.044
CW_Marxist	0.047	0.038	0.056	0.038	0.048	0.039	0.037	0.039
COL_British	−0.096	0.105	−0.097	0.106	−0.096	0.105	−0.094	0.103
COL_Portuguese	0.014	0.146	0.016	0.148	0.020	0.146	0.027	0.143
COL_Spanish	−0.356	0.291	−0.318	0.296	−0.323	0.292	−0.356	0.285
COL_Italian	−0.152	0.214	−0.134	0.218	−0.144	0.215	−0.154	0.210
COL_Belgian	−0.070	0.296	−0.071	0.301	−0.067	0.297	−0.068	0.290
COL_German	0.015	0.134	0.014	0.137	0.018	0.135	0.023	0.132
COL_US	−0.020	0.280	0.005	0.285	0.014	0.281	0.018	0.274
COL_None	−0.204	0.296	−0.216	0.301	−0.201	0.297	−0.184	0.290
COL_Unique	0.459	0.300	0.464	0.304	0.468	0.300	0.471	0.294
REG_E_Africa	0.005	0.158	0.013	0.160	0.010	0.158	0.009	0.155
REG_S_Africa	−0.120	0.169	−0.109	0.172	−0.108	0.170	−0.107	0.166
REG_C_Africa	−0.098	0.175	−0.083	0.177	−0.087	0.175	−0.091	0.171
REG_W_Africa	−0.183	0.150	−0.174	0.152	−0.168	0.150	−0.160	0.147
SD (Intercept YEAR)	0.161		0.151		0.139		0.130	
SD (Intercept CCODE)	0.263		0.268		0.264		0.258	
SD (Observations)	0.272		0.270		0.271		0.272	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 7: **ODAG IO: Reduced Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	1.241***	0.115	1.254***	0.115	1.283***	0.114	1.306***	0.115
V2X_LIBDEM	0.846***	0.064						
V2X_LIBDEM_lag1			0.808***	0.064				
V2X_LIBDEM_lag2					0.767***	0.065		
V2X_LIBDEM_lag3							0.749***	0.066
SD (Intercept YEAR)	0.825		0.824		0.816		0.809	
SD (Intercept CCODE)	0.472		0.467		0.470		0.473	
SD (Observations)	0.311		0.313		0.313		0.312	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 8: **ODAG IO: Reduced Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	1.363***	0.113	1.400***	0.113	1.438***	0.113	1.471***	0.113
POLITY2	0.006***	0.002						
POLITY2_lag1			0.005**	0.002				
POLITY2_lag2					0.005**	0.002		
POLITY2_lag3							0.004**	0.002
SD (Intercept YEAR)	0.828		0.822		0.816		0.812	
SD (Intercept CCODE)	0.392		0.397		0.406		0.412	
SD (Observations)	0.309		0.308		0.308		0.307	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 9: **ODAG IO: Reduced Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	2.150***	0.102	2.152***	0.101	2.148***	0.101	2.159***	0.100
TOTLIB1	−0.027***	0.003						
TOTLIB1_lag1			−0.025***	0.003				
TOTLIB1_lag2					−0.022***	0.003		
TOTLIB1_lag3							−0.022***	0.003
SD (Intercept YEAR)	0.513		0.498		0.487		0.475	
SD (Intercept CCODE)	0.494		0.497		0.499		0.502	
SD (Observations)	0.299		0.299		0.298		0.297	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 10: **ODAG IO: Full Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−0.142	0.267	−0.110	0.266	−0.089	0.266	−0.066	0.265
V2X_LIBDEM	0.873***	0.062						
V2X_LIBDEM_lag1			0.831***	0.063				
V2X_LIBDEM_lag2					0.784***	0.064		
V2X_LIBDEM_lag3							0.754***	0.064
logPOP	0.773***	0.055	0.775***	0.055	0.770***	0.055	0.772***	0.055
logGNI	−0.305***	0.030	−0.312***	0.031	−0.311***	0.031	−0.315***	0.032
logTRDIMtot	0.171***	0.020	0.172***	0.020	0.178***	0.020	0.180***	0.021
logTRDEXtot	−0.114***	0.019	−0.115***	0.019	−0.118***	0.019	−0.120***	0.020
COLD_WAR	−0.901***	0.119	−0.911***	0.118	−0.922***	0.118	−0.939***	0.120
CW_Socialist	−0.105**	0.034	−0.100**	0.034	−0.099**	0.035	−0.100**	0.035
CW_Marxist	0.013	0.036	0.022	0.036	0.023	0.036	0.024	0.036
COL_British	−0.179*	0.088	−0.167+	0.088	−0.158+	0.088	−0.151+	0.087
COL_Portuguese	−0.101	0.124	−0.095	0.123	−0.093	0.123	−0.090	0.122
COL_Spanish	−0.326	0.246	−0.332	0.245	−0.339	0.245	−0.338	0.242
COL_Italian	−0.278	0.182	−0.276	0.182	−0.281	0.182	−0.287	0.180
COL_Belgian	0.011	0.250	0.016	0.249	0.025	0.249	0.034	0.246
COL_German	−0.038	0.114	−0.038	0.113	−0.037	0.113	−0.035	0.112
COL_US	−0.162	0.236	−0.159	0.235	−0.150	0.234	−0.142	0.232
COL_None	0.070	0.251	0.064	0.250	0.065	0.249	0.069	0.247
COL_Unique	−0.214	0.257	−0.220	0.256	−0.223	0.256	−0.223	0.254
REG_E_Africa	0.209	0.134	0.208	0.134	0.210	0.134	0.200	0.133
REG_S_Africa	0.111	0.144	0.102	0.143	0.103	0.143	0.098	0.141
REG_C_Africa	0.151	0.148	0.144	0.148	0.143	0.147	0.131	0.146
REG_W_Africa	0.159	0.127	0.156	0.127	0.160	0.127	0.155	0.125
SD (Intercept YEAR)	0.451		0.449		0.449		0.456	
SD (Intercept CCODE)	0.222		0.221		0.220		0.218	
SD (Observations)	0.274		0.274		0.274		0.274	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 11: **ODAG IO: Full Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.377	0.262	0.363	0.259	0.300	0.262	0.272	0.260
POLITY2	0.002	0.002						
POLITY2_lag1			0.001	0.002				
POLITY2_lag2					0.002	0.002		
POLITY2_lag3							0.001	0.002
logPOP	0.730***	0.056	0.729***	0.056	0.726***	0.057	0.736***	0.057
logGNI	−0.351***	0.032	−0.344***	0.032	−0.327***	0.033	−0.329***	0.033
logTRDIMtot	0.190***	0.021	0.187***	0.021	0.193***	0.021	0.194***	0.022
logTRDEXtot	−0.112***	0.020	−0.110***	0.020	−0.115***	0.020	−0.117***	0.021
COLD_WAR	−1.021***	0.126	−1.029***	0.122	−1.028***	0.119	−1.041***	0.118
CW_Socialist	−0.125***	0.036	−0.119***	0.036	−0.103**	0.037	−0.095**	0.037
CW_Marxist	−0.022	0.038	−0.018	0.038	−0.016	0.038	−0.012	0.038
COL_British	−0.141+	0.077	−0.130+	0.077	−0.125	0.079	−0.116	0.079
COL_Portuguese	−0.094	0.115	−0.082	0.114	−0.080	0.117	−0.075	0.117
COL_Spanish	−0.477*	0.210	−0.457*	0.209	−0.463*	0.214	−0.459*	0.213
COL_Italian	−0.451**	0.156	−0.435**	0.156	−0.425**	0.159	−0.416**	0.159
COL_Belgian	0.024	0.211	0.023	0.211	0.018	0.216	0.024	0.215
COL_German	0.004	0.095	0.001	0.095	−0.005	0.097	−0.005	0.097
COL_US	−0.172	0.198	−0.172	0.197	−0.162	0.202	−0.153	0.201
COL_None	0.067	0.212	0.063	0.212	0.054	0.216	0.050	0.216
COL_Unique	−0.297	0.232	−0.297	0.227	−0.219	0.232	−0.216	0.228
REG_E_Africa	0.146	0.115	0.156	0.114	0.184	0.117	0.187	0.117
REG_S_Africa	0.118	0.125	0.108	0.125	0.117	0.127	0.118	0.127
REG_C_Africa	0.068	0.127	0.071	0.127	0.099	0.130	0.101	0.130
REG_W_Africa	0.143	0.109	0.148	0.109	0.174	0.111	0.180	0.111
SD (Intercept YEAR)	0.470		0.457		0.447		0.447	
SD (Intercept CCODE)	0.184		0.183		0.188		0.187	
SD (Observations)	0.284		0.283		0.285		0.284	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 12: **ODAG IO: Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.184	0.251	0.118	0.254	0.093	0.256	0.070	0.257
TOTLIB1	−0.031***	0.003						
TOTLIB1_lag1			−0.029***	0.003				
TOTLIB1_lag2					−0.026***	0.003		
TOTLIB1_lag3							−0.025***	0.003
logPOP	0.786***	0.055	0.781***	0.056	0.769***	0.057	0.763***	0.058
logGNI	−0.289***	0.033	−0.280***	0.034	−0.272***	0.034	−0.266***	0.034
logTRDIMtot	0.158***	0.020	0.177***	0.022	0.175***	0.022	0.178***	0.022
logTRDEXtot	−0.083***	0.020	−0.099***	0.021	−0.094***	0.021	−0.096***	0.022
COLD_WAR	−0.576***	0.071	−0.541***	0.067	−0.517***	0.064	−0.490***	0.062
CW_Socialist	−0.059	0.042	−0.064	0.042	−0.074+	0.043	−0.081+	0.044
CW_Marxist	0.120**	0.037	0.114**	0.038	0.101**	0.038	0.089*	0.038
COL_British	−0.110	0.086	−0.103	0.087	−0.094	0.088	−0.087	0.089
COL_Portuguese	−0.074	0.120	−0.064	0.122	−0.052	0.123	−0.044	0.124
COL_Spanish	−0.277	0.239	−0.261	0.243	−0.276	0.246	−0.277	0.248
COL_Italian	−0.277	0.177	−0.271	0.180	−0.279	0.182	−0.282	0.183
COL_Belgian	0.032	0.244	0.030	0.248	0.032	0.250	0.035	0.252
COL_German	0.024	0.110	0.020	0.112	0.023	0.113	0.025	0.114
COL_US	−0.124	0.230	−0.104	0.234	−0.096	0.237	−0.086	0.238
COL_None	0.024	0.244	0.014	0.248	0.019	0.250	0.026	0.252
COL_Unique	−0.042	0.251	−0.054	0.254	−0.059	0.257	−0.057	0.258
REG_E_Africa	0.215	0.131	0.234+	0.133	0.243+	0.135	0.248+	0.136
REG_S_Africa	0.109	0.140	0.123	0.142	0.126	0.144	0.131	0.145
REG_C_Africa	0.141	0.145	0.155	0.147	0.156	0.148	0.159	0.149
REG_W_Africa	0.201	0.124	0.219+	0.126	0.230+	0.128	0.238+	0.128
SD (Intercept YEAR)	0.231		0.213		0.201		0.190	
SD (Intercept CCODE)	0.215		0.219		0.221		0.223	
SD (Observations)	0.267		0.267		0.267		0.267	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 13: **ODAG Total: Reduced Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	1.662***	0.121	1.673***	0.120	1.707***	0.118	1.742***	0.115
V2X_LIBDEM	0.923***	0.066						
V2X_LIBDEM_lag1			0.922***	0.065				
V2X_LIBDEM_lag2					0.925***	0.066		
V2X_LIBDEM_lag3							0.935***	0.066
SD (Intercept YEAR)	0.870		0.860		0.828		0.792	
SD (Intercept CCODE)	0.500		0.495		0.497		0.500	
SD (Observations)	0.322		0.319		0.316		0.314	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 14: **ODAG Total: Reduced Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.652***	0.082	1.864***	0.118	1.912***	0.116	1.959***	0.113
POLITY2	−0.001	0.002						
POLITY2_lag1			0.006***	0.002				
POLITY2_lag2					0.005**	0.002		
POLITY2_lag3							0.004**	0.002
SD (Intercept YEAR)	0.495		0.858		0.830		0.797	
SD (Intercept CCODE)	0.336		0.426		0.430		0.432	
SD (Observations)	0.350		0.313		0.308		0.306	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 15: **ODAG Total: Reduced Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	2.676***	0.101	2.687***	0.100	2.678***	0.100	2.677***	0.100
TOTLIB1	−0.035***	0.003						
TOTLIB1_lag1			−0.034***	0.003				
TOTLIB1_lag2					−0.031***	0.003		
TOTLIB1_lag3							−0.030***	0.003
SD (Intercept YEAR)	0.487		0.476		0.468		0.463	
SD (Intercept CCODE)	0.510		0.511		0.510		0.511	
SD (Observations)	0.306		0.305		0.306		0.306	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 16: **ODAG Total: Full Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.051	0.292	0.011	0.294	−0.005	0.293	0.045	0.287
V2X_LIBDEM	0.935***	0.061						
V2X_LIBDEM_lag1			0.935***	0.060				
V2X_LIBDEM_lag2					0.922***	0.061		
V2X_LIBDEM_lag3							0.891***	0.061
logPOP	0.758***	0.061	0.760***	0.061	0.755***	0.061	0.752***	0.060
logGNI	−0.209***	0.030	−0.211***	0.030	−0.208***	0.030	−0.218***	0.030
logTRDIMtot	0.145***	0.019	0.156***	0.019	0.166***	0.019	0.156***	0.020
logTRDEXtot	−0.061***	0.018	−0.064***	0.018	−0.067***	0.018	−0.052**	0.019
COLD_WAR	−0.560***	0.092	−0.548***	0.088	−0.534***	0.084	−0.537***	0.082
CW_Socialist	−0.030	0.033	−0.011	0.033	−0.006	0.033	−0.004	0.033
CW_Marxist	0.025	0.035	0.025	0.034	0.018	0.034	0.012	0.034
COL_British	−0.118	0.103	−0.119	0.104	−0.122	0.104	−0.123	0.102
COL_Portuguese	0.016	0.143	0.019	0.145	0.016	0.145	0.016	0.142
COL_Spanish	−0.304	0.286	−0.308	0.290	−0.320	0.290	−0.339	0.283
COL_Italian	−0.213	0.211	−0.193	0.213	−0.187	0.213	−0.188	0.209
COL_Belgian	0.045	0.291	0.022	0.295	0.004	0.295	−0.013	0.288
COL_German	−0.029	0.132	−0.035	0.134	−0.042	0.134	−0.040	0.131
COL_US	0.064	0.275	0.064	0.278	0.060	0.278	0.043	0.272
COL_None	−0.108	0.291	−0.116	0.295	−0.113	0.295	−0.099	0.288
COL_Unique	0.105	0.295	0.099	0.298	0.097	0.298	0.101	0.291
REG_E_Africa	0.023	0.155	0.044	0.157	0.056	0.157	0.050	0.154
REG_S_Africa	−0.137	0.166	−0.118	0.169	−0.100	0.169	−0.100	0.165
REG_C_Africa	−0.070	0.171	−0.045	0.174	−0.025	0.174	−0.027	0.170
REG_W_Africa	−0.143	0.147	−0.123	0.149	−0.104	0.149	−0.101	0.146
SD (Intercept YEAR)	0.339		0.326		0.310		0.300	
SD (Intercept CCODE)	0.259		0.263		0.263		0.256	
SD (Observations)	0.266		0.262		0.260		0.260	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 17: **ODAG Total: Full Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.614*	0.310	0.522+	0.305	0.485	0.301	0.532+	0.291
POLITY2	0.006***	0.002						
POLITY2_lag1			0.005***	0.002				
POLITY2_lag2					0.005***	0.002		
POLITY2_lag3							0.004*	0.002
logPOP	0.679***	0.068	0.688***	0.067	0.676***	0.066	0.671***	0.064
logGNI	−0.217***	0.032	−0.217***	0.032	−0.206***	0.032	−0.218***	0.031
logTRDIMtot	0.152***	0.021	0.162***	0.021	0.176***	0.021	0.166***	0.021
logTRDEXtot	−0.050*	0.020	−0.050*	0.020	−0.055**	0.020	−0.039+	0.020
COLD_WAR	−0.677***	0.097	−0.662***	0.090	−0.641***	0.083	−0.642***	0.079
CW_Socialist	−0.037	0.036	−0.016	0.035	0.011	0.035	0.025	0.035
CW_Marxist	0.001	0.037	−0.003	0.037	−0.009	0.036	−0.014	0.036
COL_British	−0.088	0.099	−0.092	0.099	−0.092	0.098	−0.083	0.095
COL_Portuguese	0.028	0.147	0.027	0.147	0.018	0.146	0.018	0.142
COL_Spanish	−0.447+	0.270	−0.436	0.269	−0.463+	0.267	−0.494+	0.259
COL_Italian	−0.383+	0.197	−0.352+	0.197	−0.338+	0.196	−0.333+	0.190
COL_Belgian	0.071	0.272	0.033	0.271	0.013	0.270	0.003	0.262
COL_German	0.025	0.123	0.013	0.122	0.005	0.122	0.008	0.118
COL_US	0.060	0.254	0.051	0.254	0.039	0.253	0.013	0.245
COL_None	−0.092	0.272	−0.107	0.271	−0.101	0.270	−0.080	0.262
COL_Unique	0.055	0.284	0.055	0.279	0.108	0.278	0.129	0.268
REG_E_Africa	−0.056	0.146	−0.018	0.145	0.011	0.144	0.015	0.140
REG_S_Africa	−0.150	0.159	−0.117	0.159	−0.091	0.158	−0.090	0.153
REG_C_Africa	−0.158	0.163	−0.117	0.162	−0.079	0.161	−0.075	0.156
REG_W_Africa	−0.155	0.139	−0.118	0.139	−0.084	0.138	−0.075	0.134
SD (Intercept YEAR)	0.351		0.326		0.300		0.284	
SD (Intercept CCODE)	0.239		0.239		0.238		0.230	
SD (Observations)	0.280		0.275		0.272		0.271	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 18: **ODAG Total: Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.582*	0.276	0.501+	0.279	0.501+	0.276	0.500+	0.271
TOTLIB1	−0.037***	0.003						
TOTLIB1_lag1			−0.037***	0.003				
TOTLIB1_lag2					−0.034***	0.003		
TOTLIB1_lag3							−0.032***	0.003
logPOP	0.778***	0.061	0.782***	0.062	0.767***	0.061	0.754***	0.061
logGNI	−0.265***	0.032	−0.257***	0.033	−0.249***	0.033	−0.243***	0.034
logTRDIMtot	0.145***	0.020	0.176***	0.021	0.174***	0.022	0.168***	0.022
logTRDEXtot	−0.011	0.019	−0.038+	0.020	−0.035+	0.021	−0.028	0.021
COLD_WAR	−0.354***	0.060	−0.318***	0.057	−0.297***	0.054	−0.279***	0.052
CW_Socialist	0.014	0.041	0.022	0.041	0.013	0.042	0.006	0.043
CW_Marxist	0.081*	0.036	0.085*	0.037	0.074*	0.037	0.062+	0.037
COL_British	−0.094	0.098	−0.093	0.100	−0.089	0.099	−0.085	0.097
COL_Portuguese	−0.020	0.137	−0.015	0.139	−0.008	0.137	0.000	0.135
COL_Spanish	−0.349	0.274	−0.310	0.278	−0.321	0.274	−0.347	0.269
COL_Italian	−0.190	0.201	−0.173	0.205	−0.184	0.202	−0.192	0.198
COL_Belgian	−0.078	0.279	−0.083	0.283	−0.080	0.279	−0.081	0.274
COL_German	0.003	0.126	−0.001	0.128	0.003	0.127	0.007	0.124
COL_US	−0.080	0.263	−0.051	0.268	−0.043	0.264	−0.037	0.259
COL_None	−0.134	0.278	−0.150	0.283	−0.136	0.279	−0.121	0.273
COL_Unique	0.271	0.282	0.268	0.287	0.268	0.283	0.271	0.278
REG_E_Africa	0.063	0.149	0.078	0.151	0.079	0.149	0.080	0.146
REG_S_Africa	−0.074	0.159	−0.056	0.162	−0.053	0.160	−0.050	0.157
REG_C_Africa	−0.003	0.164	0.019	0.167	0.016	0.165	0.015	0.162
REG_W_Africa	−0.065	0.141	−0.048	0.144	−0.039	0.142	−0.029	0.139
SD (Intercept YEAR)	0.190		0.175		0.162		0.152	
SD (Intercept CCODE)	0.247		0.252		0.248		0.243	
SD (Observations)	0.260		0.259		0.260		0.261	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001